

Production Network Structure and Economic Outcomes: Evidence from Input-Output Data

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April 2026

Abstract

This dissertation investigates how production networks shape the pass-through of oil shocks into consumer prices. Using OECD Inter-Country Input-Output Tables covering 80 economies and 50 industries from 1995 to 2022, I construct a sectoral exposure measure and a country-level multiplier from a Leontief cost-push framework. I then estimate local projections of quarterly CPI inflation on exposure-weighted oil shocks for Eurozone countries. The country-level multiplier ξ fails to explain within-quarter cross-country heterogeneity in CPI responses to oil shocks once time fixed effects are introduced, but captures the average scale of oil-shock pass-through in specifications without time fixed effects.

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7494 words (including 1700 for diagrams, equations, tables)

1 Introduction

In 2022, a series of supply shocks hit the global economy. An avian flu outbreak in the United States caused egg prices to sharply rise, while the Russo-Ukrainian war drove a surge in global energy prices. Both became symbols of a broader cost-of-living crisis. Yet, few worried that soaring egg prices would lead to broad-based inflation, whereas surging oil prices immediately revived stagflation fears reminiscent of the 1970s (Reuters 2022). People intuitively understand that energy’s deep embedding in production networks means that energy price shocks carry greater weight for the aggregate price level.

Long-run trend inflation is a monetary phenomenon. The Great Inflation in the 1970s was not primarily driven by oil supply shocks (Barsky & Kilian 2004); rather, it was the product of loose monetary policy following the collapse of Bretton Woods. Today, with independent central banks and anchored inflation expectations, it is hard to imagine a repeat of the 1970s. However, central banks operating under dual mandates cannot fully offset cost-push inflation. Freed from golden fetters, central banks are still bound by oil shackles. For this reason, understanding how energy shocks pass through to consumer prices remains important.

Existing work documents that oil shocks pass through into headline inflation with cross-country heterogeneity (Abdallah & Kpodar 2023), and that input-output linkages shape the transmission of cost shocks to prices (Alvarez & Kroen 2025). Oil shocks raise the cost of inputs of directly-exposed sectors; these higher costs are then passed to downstream sectors through further rounds of propagation, and finally to consumer prices. However, no existing measure connects production network structure to a country-level prediction of how much a given economy’s CPI should respond to a common energy shock.

This dissertation constructs such measures. I develop sectoral and country-level measures of oil exposure embedded in an economy’s cost structure and consumption basket. The framework is general since any sector may be designated as the exogenous source of cost shocks. To test whether the exposure measures predict pass-through, I relate the country-level multiplier to quarterly CPI inflation using local projections. The empirical analysis focuses on Eurozone countries as the baseline sample to reduce heterogeneity in monetary policy regimes, and then considers instrumental variables and alternative sample extensions.

I find that the country-level multiplier captures the magnitude of oil-shock pass-through but fails to explain its cross-country heterogeneity. In specifications without time fixed effects, exposure-weighted oil shocks enter positively, with a statistically significant hump-shaped profile across horizons. The estimated magnitudes are consistent with the pre-

dicted scale of pass-through and with existing estimates by Choi et al. (2018). However, once time fixed effects are introduced, the relationship becomes statistically insignificant.

The rest of the paper is structured as follows: Section 2 reviews the related literature on production networks, energy-shock pass-through, and input-output approaches to inflation. Section 3 develops the Leontief cost-push framework and defines the sectoral and country-level exposure measures. Section 4 describes the data and measurement choices. Section 5 documents the properties of the constructed exposure measures. Section 6 sets out the empirical strategy. Section 7 presents the results, and Section 8 discusses their interpretation and limitations.

2 Literature Review

The notion that economic outcomes are shaped by production networks is not new to economics, dating back to at least the 1930s. The classic input-output framework of Leontief (1936, 1941) provides a powerful tool for studying the interdependence within production networks, on which this dissertation builds. In particular, this dissertation relates to three strands of literature: production networks, energy-shock pass-through, and input-output approaches to inflation transmission.

Within the first strand, the production-network literature shows that sectoral shocks can have aggregate consequences. Hulten (1978) gives a simple first-order perspective, motivated by the envelope theorem, where the aggregate effects of sectoral shocks are captured by a sufficient statistic, the Domar weight. When there is significant asymmetry in the roles that sectors play as suppliers, idiosyncratic sectoral shocks can become the origins of aggregate fluctuations (Acemoglu et al. 2012). This is supported by empirical work at both the sectoral and firm levels. Acemoglu et al. (2016) document that sectoral shocks propagate through input-output and geographic linkages in U.S. data, while Carvalho et al. (2021) show that localised disruptions from the Great East Japan Earthquake propagated both upstream and downstream through firm-level supply chains. Additionally, Baqaee & Farhi (2019) go beyond first-order approximations and show that non-linearities can be large depending on the structure of the production network.

This dissertation takes a narrower approach. Similar to the first-order tradition, I seek a tractable exposure measure, but apply it to price transmission rather than aggregate output.

Empirical work on energy shocks and inflation generally finds that energy price shocks do pass through into headline inflation. Choi et al. (2018) provide evidence that oil price

shocks contribute to inflation in both advanced and developing economies. Abdallah & Kpodar (2023) show that there is significant cross-country heterogeneity in the response of domestic inflation to retail energy price shocks. More recently, Alvarez & Kroen (2025) argue that energy price shocks played a central role in the post-pandemic global inflation surge, while showing that sectoral linkages have additional explanatory power for understanding pass-through. The gap that this dissertation addresses is whether production network structure can predict the magnitude and heterogeneity of pass-through.

There is also theoretical work linking inflation transmission and input-output linkages. Nordhaus & Shoven (1977) use input-output matrices to decompose inflation into exogenous and endogenous sectoral components. Recent work by Afrouzi & Bhattarai (2023) shows that production linkages amplify the persistence of inflation in response to monetary shocks in a multi-sector New Keynesian model. Empirically, Przybyliński & Gorzalczyński (2022) apply a Leontief price model to Danish data and decompose inflation into underlying cost channels, principally value added and import prices.

This dissertation synthesises the energy shock pass-through literature with the input-output transmission literature. Rather than decomposing realised inflation into contemporaneous cost components, I use input-output data to construct ex-ante measures by seeding exogenous sectors, then test whether the measure predicts heterogeneous pass-through in a large cross-country panel.

3 Theoretical Framework

3.1 Leontief Cost-Push Price System

I begin with a benchmark Leontief cost-push price system. Assume that production is characterised by fixed input requirements in the short run and that firms earn zero profits. Then the price of sector j 's output equals the cost of intermediate inputs plus primary inputs per unit of output.

Denote technical coefficients $\alpha_{ij} \equiv \frac{q_{ij}}{q_j}$, representing the physical quantity of input i to produce one unit of output in sector j . q_{ij} is the quantity of input i used by sector j , q_j is sector j 's output quantity, and p_j its price. Then sectoral unit costs satisfy

$$p_j q_j = \left(\sum_{i=1}^n p_i q_{ij} \right) + v_j,$$

where v_j denotes value added in sector j , including indirect taxes less subsidies.

Dividing through by q_j gives

$$p_j = (\sum_{i=1}^n p_i \alpha_{ij}) + \frac{v_j}{q_j}$$

Input-output tables, however, are observed in nominal values rather than physical quantities. The nominal technical coefficients a_{ij} are therefore

$$a_{ij} \equiv \frac{p_i q_{ij}}{p_j q_j} = \frac{p_i}{p_j} \alpha_{ij}$$

Under the standard normalisation in which all prices are set equal to one in the base year (Miller & Blair 2022), monetary and physical coefficients coincide, so $a_{ij} = \alpha_{ij}$ at the benchmark. Consequently, prices do not represent real prices but index prices that have a value of one in the base year (Oosterhaven 2024). Under such an assumption, the matrix of the technical coefficients in monetary units can be treated like the matrix of technical coefficients in physical units.

Denote $c_j \equiv \frac{v_j}{q_j}$, representing the primary-input cost per unit of output in sector j .

$$p_j = (\sum_{i=1}^n a_{ij} p_i) + c_j$$

In vector form:

$$\mathbf{p} = \mathbf{A}^\top \mathbf{p} + \mathbf{c}, \quad \mathbf{p} = (\mathbf{I} - \mathbf{A}^\top)^{-1} \mathbf{c}$$

Note that $(\mathbf{I} - \mathbf{A}^\top)^{-1}$ is the transpose of the standard Leontief inverse in input-output literature.

3.2 Exogenous-Endogenous Partition

I partition the economy into non-exogenous sectors N and exogenous sectors E (Nordhaus & Shoven 1977).

$$\mathbf{p} = \begin{pmatrix} \mathbf{p}_N \\ \mathbf{p}_E \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} \mathbf{A}_{NN} & \mathbf{A}_{NE} \\ \mathbf{A}_{EN} & \mathbf{A}_{EE} \end{pmatrix} \quad \mathbf{A}^\top = \begin{pmatrix} \mathbf{A}_{NN}^\top & \mathbf{A}_{EN}^\top \\ \mathbf{A}_{NE}^\top & \mathbf{A}_{EE}^\top \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} \mathbf{c}_N \\ \mathbf{c}_E \end{pmatrix}$$

Starting from the benchmark Leontief price system,

$$\mathbf{p} = \mathbf{A}^\top \mathbf{p} + \mathbf{c},$$

Index prices of non-exogenous sectors are as follows:

$$\mathbf{p}_N = \mathbf{A}_{NN}^\top \mathbf{p}_N + \mathbf{A}_{EN}^\top \mathbf{p}_E + \mathbf{c}_N$$

Hence,

$$\mathbf{p}_N = (\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1}(\mathbf{A}_{EN}^\top \mathbf{p}_E + \mathbf{c}_N)$$

The above expression makes it clear that prices in non-exogenous sectors depend on both exogenous inputs $\mathbf{p}_E$ and primary inputs $\mathbf{c}_N$. The cost of exogenous inputs is transmitted directly through $\mathbf{A}_{EN}^\top$. Both are then transmitted indirectly through $(\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1}$.

3.3 Growth-Rate Representation

To study inflation, it is more convenient to rewrite the benchmark price system in growth rates instead. Let hats denote growth rates, so that for any variable x :

$$\hat{x} \equiv \frac{1}{x} \frac{dx}{dt}$$

Using $\mathbf{p}_N = \mathbf{A}_{NN}^\top \mathbf{p}_N + \mathbf{A}_{EN}^\top \mathbf{p}_E + \mathbf{c}_N$ and differentiating while holding $\mathbf{A}$ fixed:

$$d\mathbf{p}_N = \mathbf{A}_{NN}^\top d\mathbf{p}_N + \mathbf{A}_{EN}^\top d\mathbf{p}_E + d\mathbf{c}_N$$

Converting to growth rates, where

$$d\mathbf{p}_N = \text{diag}(\mathbf{p}_N) \hat{\mathbf{p}}_N \quad d\mathbf{p}_E = \text{diag}(\mathbf{p}_E) \hat{\mathbf{p}}_E \quad d\mathbf{c}_N = \text{diag}(\mathbf{c}_N) \hat{\mathbf{c}}_N$$

Evaluating again where the prices are normalised to one, this becomes

$$\hat{\mathbf{p}}_N = \mathbf{A}_{NN}^\top \hat{\mathbf{p}}_N + \mathbf{A}_{EN}^\top \hat{\mathbf{p}}_E + \text{diag}(\mathbf{s}_N) \hat{\mathbf{c}}_N,$$

where $\mathbf{s} \equiv [\frac{v_i}{p_i q_i}] = [\frac{c_i}{p_i}]$, representing the vector of value-added shares.

Thus,

$$\hat{\mathbf{p}}_N = (\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1} \mathbf{A}_{EN}^\top \hat{\mathbf{p}}_E + (\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1} (\text{diag}(\mathbf{s}_N) \hat{\mathbf{c}}_N)$$

3.4 Sectoral Seeded Exposure τ

I now define the sector-level exposure of non-exogenous prices to a common exogenous shock. Let u denote a scalar exogenous shock, such as an oil supply news shock, and assume it loads onto the exogenous sectors according to a vector of price loadings $\boldsymbol{\lambda}$

$$\hat{\mathbf{p}}_E = \boldsymbol{\lambda} u$$

The sectoral seeded energy exposure vector is defined as

$$\boldsymbol{\tau} \equiv (\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1} \mathbf{A}_{EN}^\top \boldsymbol{\lambda}$$

Mathematically, $\boldsymbol{\tau}$ can be viewed as a generalised Katz-Bonacich centrality on the non-exogenous network (Katz 1953, Bonacich 1987), with heterogeneous seeds given by direct exogenous exposure $\mathbf{A}_{EN}^\top \boldsymbol{\lambda}$. Related Bonacich-Katz centrality measures have been applied to input-output data by Criscuolo & Timmis (2018), who use them to map central hubs in global value chains. Their finding that petroleum-related country-sector nodes are relatively central upstream suppliers is consistent with treating energy sectors as plausible seeded sources of downstream cost propagation.

The non-exogenous price system can be written compactly as

$$\hat{\mathbf{p}}_N = \boldsymbol{\tau} u + (\mathbf{I} - \mathbf{A}_{NN}^\top)^{-1} (\text{diag}(\mathbf{s}_N) \hat{\mathbf{c}}_N)$$

Each component τ_i measures the total exposure of non-exogenous sector i to the common shock. It captures both direct purchases and indirect exposures through upstream suppliers.

Expanding the inverse as a Neumann series clarifies the interpretation:

$$\boldsymbol{\tau} = \mathbf{A}_{EN}^\top \boldsymbol{\lambda} + \sum_{k=1}^{\infty} (\mathbf{A}_{NN}^\top)^k \mathbf{A}_{EN}^\top \boldsymbol{\lambda}$$

It illustrates that $\boldsymbol{\tau}$ captures direct exposure to seeded exogenous sectors through the first term, while the second term captures higher-order rounds of cost propagation along production chains of increasing lengths through successive powers of $\mathbf{A}_{NN}^\top$. Thus, a sector may have a high τ because it buys directly from exogenous sectors, or because it relies on suppliers whose own costs are heavily exposed to exogenous sectors.

As such, this gives a natural decomposition into direct and indirect components:

$$\boldsymbol{\tau}_{\text{dir}} \equiv \mathbf{A}_{EN}^\top \boldsymbol{\lambda} \quad \boldsymbol{\tau}_{\text{ind}} \equiv \sum_{k=1}^{\infty} (\mathbf{A}_{NN}^\top)^k \mathbf{A}_{EN}^\top \boldsymbol{\lambda}$$

The recursive interpretation

$$(\mathbf{I} - \mathbf{A}_{NN}^\top) \boldsymbol{\tau} = \mathbf{A}_{EN}^\top \boldsymbol{\lambda} \implies \boldsymbol{\tau} = \underbrace{\mathbf{A}_{EN}^\top \boldsymbol{\lambda}}_{\text{direct}} + \underbrace{\mathbf{A}_{NN}^\top \boldsymbol{\tau}}_{\text{indirect}}$$

also demonstrates this decomposition.

This recursive form is analogous to trophic structure in ecology (Farmer 2024). In a food web, a predator can have a high trophic level if it consumes many species that are themselves high on the food web. Similarly, a sector may be heavily exposed to exogenous

shocks because its inputs are sourced from suppliers that depend heavily on the exogenous sectors. A standard formulation of trophic levels takes the recursive form above, using $\mathbf{t} = \mathbf{1} + \mathbf{A}^\top \mathbf{t}$, where the vector $\mathbf{t}$ is the vector of trophic levels, $\mathbf{A}$ is the matrix $[a_{ij}]$, the fraction of species i in the diet of j (Levine 1980).

3.5 Country-Level Multiplier ξ

The next step is to aggregate these sectoral exposures into a country-level measure of inflation exposure using consumer price index (CPI) expenditure weights. Let $\mathbf{w}_{m,t}$ denote the vector of CPI expenditure weights on each sector in country m .

CPI inflation can be approximated as

$$\pi_m \approx \mathbf{w}_m^\top \hat{\mathbf{p}}_m$$

Substitute:

$$\pi_m \approx \mathbf{w}_{E,m}^\top \boldsymbol{\lambda}_m u + \mathbf{w}_{N,m}^\top \boldsymbol{\tau}_m u + \mathbf{w}_{N,m}^\top (\mathbf{I} - \mathbf{A}_{NN,m}^\top)^{-1} (\text{diag}(\mathbf{s}_{N,m}) \hat{\mathbf{c}}_{N,m})$$

The country-level multiplier is defined as

$$\xi_m \equiv \underbrace{\mathbf{w}_{E,m}^\top \boldsymbol{\lambda}_m}_{\text{direct}} + \underbrace{\mathbf{w}_{N,m}^\top \boldsymbol{\tau}_m}_{\text{indirect}}$$

The seeded energy multiplier ξ_m measures both the direct and indirect energy shares embedded in a unit of household consumption in country m , as implied by the Leontief cost-push price system.

Similar to $\boldsymbol{\tau}$, ξ can be decomposed into a direct and two indirect components:

$$\xi_{\text{dir},m} \equiv \mathbf{w}_{E,m}^\top \boldsymbol{\lambda}_m \quad \xi_{\text{near},m} \equiv \mathbf{w}_{N,m}^\top \boldsymbol{\tau}_{\text{dir},m} \quad \xi_{\text{deep},m} \equiv \mathbf{w}_{N,m}^\top \boldsymbol{\tau}_{\text{ind},m}$$

$\xi_{\text{dir},m}$ captures the direct exposure of households to exogenous sectors. $\xi_{\text{near},m}$ captures the first indirect layer of exposure, where shocks are passed to households through non-exogenous sectors. $\xi_{\text{deep},m}$ captures higher-order indirect exposure through all longer production chains.

With ξ_m , inflation can be written compactly as:

$$\pi_m \approx \xi_m u + z_m$$

where $z_m \equiv \mathbf{w}_{N,m}^\top (\mathbf{I} - \mathbf{A}_{NN,m}^\top)^{-1} (\text{diag}(\mathbf{s}_{N,m}) \hat{\mathbf{c}}_{N,m})$

Countries with higher ξ_m are therefore predicted to exhibit stronger inflationary responses to a common global energy shock, *ceteris paribus*.

3.6 Testable Prediction and Empirical Scope

The theoretical framework yields a simple empirical prediction. Holding other determinants of inflation fixed, if two countries are hit by the same shock u , the model predicts a larger inflation response in the country with the higher value of ξ_m .

The theory is most likely to hold in the short run. Over short horizons, production technologies, input-output linkages, and expenditure patterns are unlikely to adjust fully. Under these conditions, the Leontief cost-push price system provides a reasonable approximation for the transmission of exogenous shocks through the production network.

As such, the scope of this framework is limited. By construction, it abstracts from endogenous substitution across inputs, changes in markups, wage-setting behaviour, and systematic monetary-policy responses. Accordingly, the mapping from exposure to observed inflation outcomes should not be interpreted as structurally invariant in the Lucas (1976) sense.

3.7 Constant Elasticity of Substitution (CES) Microfoundation

Although the benchmark analysis above is conducted in a Leontief framework, the central exposure objects can also be derived from a richer CES environment. Around the benchmark allocation, the first-order response of non-energy prices to a common energy shock takes the same network-weighted form as the seeded exposure measures τ_m and ξ_m derived above. The baseline model is best understood as a transparent short-run approximation to a more general production structure, rather than a comprehensive theory of supply-driven inflation.

The advantage of the richer microfoundation is that it allows for endogenous substitution and gives rise to additional higher-order terms capturing reallocation effects. These terms are not the focus of the baseline empirical analysis, which is built around the first-order benchmark prediction summarized by ξ_m . I therefore keep the main text focused on the Leontief benchmark and present the full CES derivation, together with the higher-order extension, in Appendix A.

4 Data and Measurement

4.1 Input-Output Data

Technical coefficient matrices $\mathbf{A}$ and sectoral value-added data $\mathbf{v}$ were taken from the OECD Inter-Country Input-Output Tables (ICIO), regular edition. The 2025 release covers 80 economies, plus a rest-of-the-world aggregate, and 50 industries annually from 1995 to 2022.

I classified two sectors as exogenous: "Extraction of Crude Petroleum and Natural Gas", and "Manufacture of Coke and Refined Petroleum Products" since the exogenous shock of interest is an oil price shock. Oil prices are primarily determined on world markets and are therefore plausibly exogenous to the domestic production structure of any individual country.

I do not directly observe $\hat{\mathbf{c}}_{N,m}$. As $c_j \equiv \frac{v_j}{q_j}$, it follows that $\hat{\mathbf{c}}_{N,m} = \hat{\mathbf{v}}_{N,m} - \hat{\mathbf{q}}_{N,m}$. However, sectoral value-added growth $\hat{\mathbf{v}}_{N,m}$ is only available at annual frequency, so I constructed a quarterly proxy by allocating annual $\hat{\mathbf{v}}_{N,m}$ evenly across quarters under a log-linear approximation. Sectoral measures of real output growth are also unavailable, so I used quarterly real GDP growth as a proxy. This produces an imperfect quarterly proxy for $\hat{\mathbf{c}}_{N,m}$, which is used only in additional specifications.

For simplicity, I set the vector of price loadings $\boldsymbol{\lambda}$ as $\mathbf{1}$, so that a unit oil shock is assumed to affect both exogenous sectors equally.

4.2 Inflation and Oil Price Data

Quarterly consumer price inflation was sourced from the International Monetary Fund (IMF) Consumer Price Index database. I took log differences of its seasonally adjusted CPI series. Taiwan was an exception, for which I used the national statistical source, as IMF coverage was unavailable. I did not use sectoral price indices in my analysis, since internationally comparable sector-level price series were too sparse across countries and industries to support a panel.

Quarterly oil-price changes were constructed from the World Bank Commodity Price Data (The Pink Sheet). I took logs of the monthly crude oil series — an average of Brent, Dubai, and WTI, then averaged within each quarter, and first-differenced the resulting quarterly log price to obtain the quarterly oil shock series. The series runs from 1985 to 2024.

I additionally used the exogenous oil supply news shock series from Känzig (2021), which uses high-frequency oil futures price movements to identify exogenous price movements. The series has been continually updated and spans 1975–2025.

Finally, I proxied CPI basket weights $\mathbf{w}$ with household final consumption expenditure (HFCE) shares from the OECD ICIO tables.

4.3 Control Variables

Exchange rates are obtained primarily from the IMF exchange-rate database (domestic currency per US dollar, period average), with supplementary sources used for Russia (Central Bank of Russia) and Myanmar (exchange-rate.org). I computed quarterly log differences for my exchange rate series.

Policy rates are taken from the Bank for International Settlements (BIS) central bank policy rate dataset. For euro-area countries, I use the common euro-area policy rate from the date of euro adoption onward.

Quarterly real GDP is drawn primarily from the IMF Quarterly National Accounts database and supplemented with official national sources where IMF coverage is absent. Country-specific supplements are used for Angola, the UAE, Belarus, China, Egypt, Malaysia, Peru, Russia, and Taiwan. I applied an HP filter (Hodrick & Prescott 1997) with smoothing parameter $\lambda = 1600$ to log real GDP and used the cyclical component as the output gap series.

4.4 Missing Data

The baseline Eurozone sample is an unbalanced panel by construction since euro adoption is staggered across countries. Within this sample, quarterly CPI inflation and all controls were fully observed, yielding 1476 country-quarter observations for 19 member countries observed between 1999Q1 and 2022Q4.

For the broader sample of 80 countries, quarterly CPI inflation was observed for 7522 country-quarters, with complete coverage for 44 countries from 1995Q1 to 2022Q4. Requiring the full control set reduces the usable sample to 4282 observations. Conditional on CPI availability, 581 observations were missing only the output gap, 1450 only the policy rate, and 1209 both variables. No observations were lost because of missing exchange-rate data. Missingness was concentrated among developing economies.

All specifications use complete-case estimation. I did not impute missing controls, since doing so would require additional assumptions on quarterly dynamics and risk introducing model-dependent variation into the regressors. If missingness is not at random, complete-case estimation on the broad sample yields inconsistent estimates. However, this concern does not apply to the Eurozone subsample, where the panel is fully observed conditional on membership.

5 Properties of the Exposure Measures

This section shows that the constructed exposure measures vary across sectors and countries in economically interpretable ways.

I begin with sectoral exposure τ , showing that it is consistent with economic intuition about energy use and production structures.

I then turn to the country-level multiplier ξ to show that it differs meaningfully between countries, providing sufficient heterogeneity to study how common energy shocks map onto different CPI inflation responses.

Finally, I show that countries differ not only in total seeded exposure, but also in composition. This motivates an empirical specification that examines whether the composition of ξ matters beyond the total level, which is explored further in Appendix B.

5.1 Sectoral Patterns in τ

Figure 1 shows that manufacturing sectors, such as steel manufacturing (STL), tend to have higher median indirect exposure τ_{ind} than services like real estate activities (REA). This pattern is consistent with the findings of McNerney et al. (2022), who show that sectors located in longer and more intermediate-input-intensive production chains inherit more of the productivity gains originating upstream in the network. In my setting, the analogous mechanism is that such sectors inherit greater upstream cost pressure.

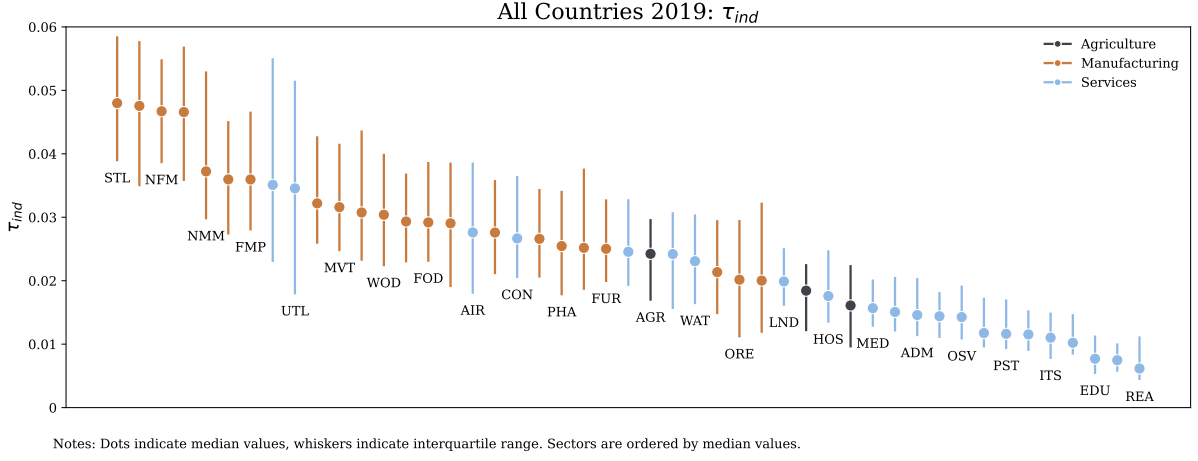


Figure 1: Indirect energy exposure τ_{ind} across sectors, all countries, 2019.

This broad sectoral pattern holds across countries. Figure 2 plots the sectoral exposures τ for China and the United States in 2019.

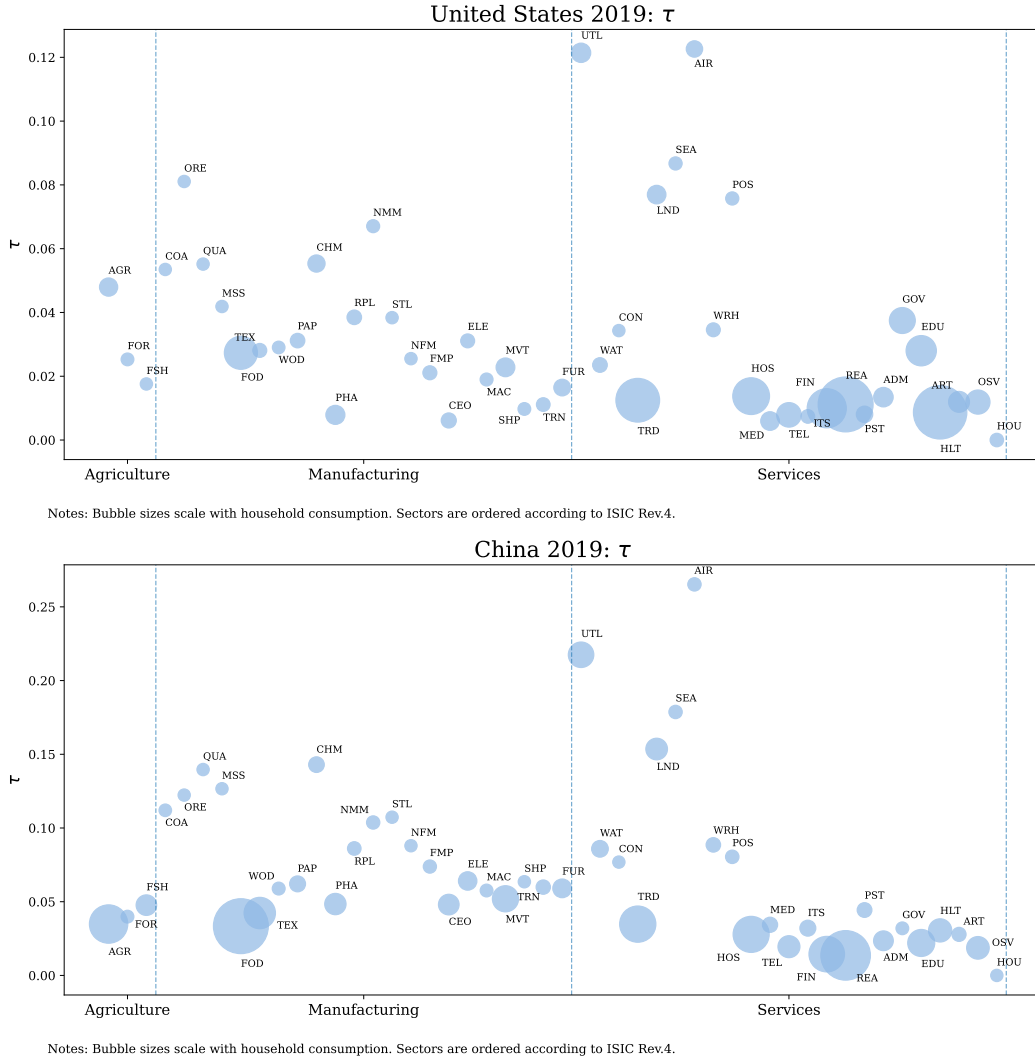


Figure 2: Sectoral seeded energy exposure τ in the United States and China, 2019.

The charts reveal that energy-intensive sectors such as utilities (UTL), chemicals (CHM), and transport (AIR/ SEA/ LND) sit consistently at the upper end of the distribution. By contrast, consumer services tend to have very low sectoral energy exposure.

The overall level of exposure differs markedly across economies. Chinese sectors are shifted upward relative to their US counterparts, suggesting that differences in sectoral exposure aggregate into meaningful cross-country differences in ξ .

5.2 Cross-Country Heterogeneity in ξ

Indeed, there is substantial cross-country heterogeneity in the country-level seeded multiplier ξ ; countries are not uniformly exposed ex ante to common energy shocks. Among Eurozone countries, the median ξ_m is 0.0334 with a standard deviation of 0.0201. Across the full sample, the median is 0.0462, which closely aligns with the aggregate oil shock pass-through coefficient of 0.043 estimated by Choi et al. (2018).

Figure 3 plots ξ over time for a selected set of economies.

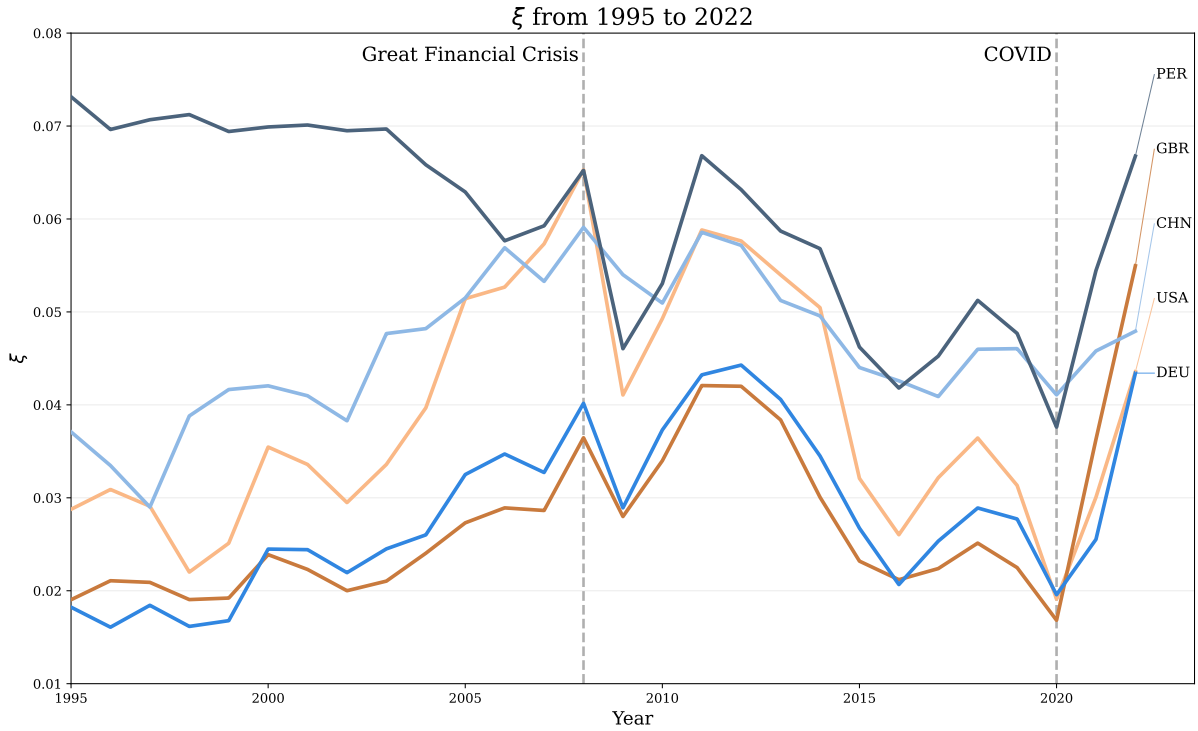


Figure 3: Country-level multiplier ξ for Peru, the UK, China, USA, and Germany from 1995 to 2022.

As shown, ξ is not static over time, with changes driven by both input-output structure and CPI expenditure weights. Even so, this variation is gradual rather than volatile,

barring large global shocks such as the Great Financial Crisis in 2008 and the COVID-19 pandemic. The cross-country ordering remains reasonably persistent as well. As such, ξ is best understood as a slow-moving exposure measure that is imperfect but economically interpretable.

As ξ evolves over time, the empirical specification should use the lagged values of ξ , so that measured exposure reflects pre-existing production networks, rather than contemporaneous adjustments to shocks.

5.3 Decomposition of ξ

The decomposition reveals that countries differ not only in the level of ξ , but also in composition. Figure 4 plots the three-way decomposition (ξ_{dir} , ξ_{near} , ξ_{deep}) for a selected set of countries.

China (CHN) is dominated by ξ_{deep} , reflecting the extraordinary depth of its production chains. Countries such as the UK (GBR) and Peru (PER) have high ξ_{near} , suggesting that they have relatively short production chains but energy-intensive production networks. Angola (AGO) presents the most extreme case; it has the highest ξ but the lowest ξ_{deep} share.

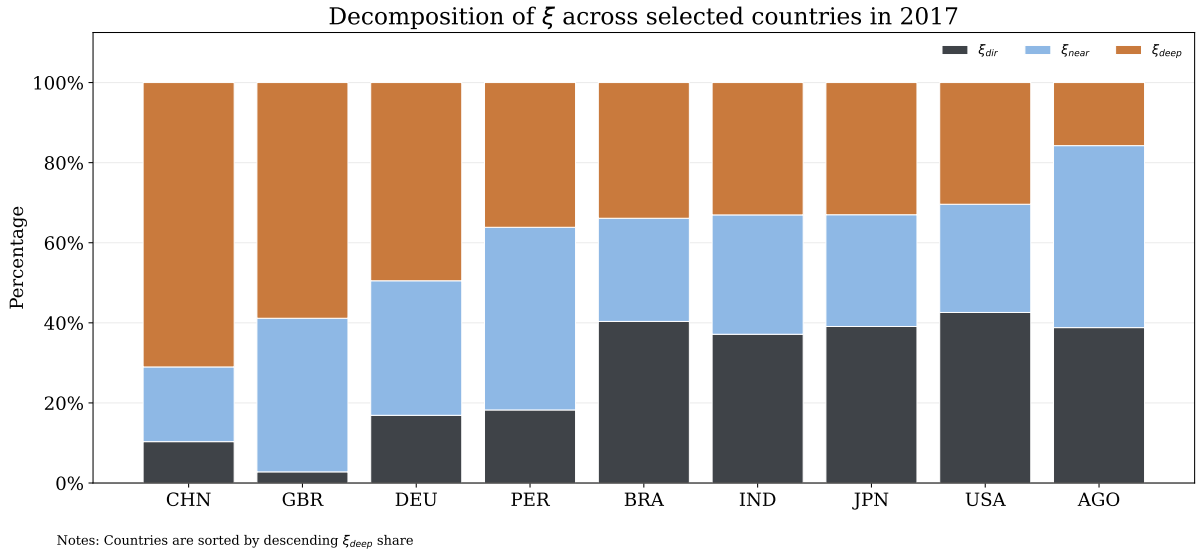


Figure 4: Decomposition of ξ for China, the UK, Germany, Peru, Brazil, India, Japan, USA, and Angola in 2017

6 Empirical Strategy

The analysis in this section is organised around one central question: do countries with larger multipliers ξ_m exhibit larger CPI pass-through in response to oil shocks u_t ?

To trace the dynamic response of inflation, I estimate local projections (Jordà 2005) up to 12 quarters ahead, using cumulative CPI inflation as the dependent variable. Local projections allow the inflation response to vary flexibly horizon by horizon without needing to impose a common autoregressive structure across countries.

For macroeconomic controls, I follow Alvarez & Kroen (2025) and include nominal exchange rate changes, the output gap, and policy rates. These variables may affect observed inflation independently of the energy-exposure channel, introducing omitted-variable bias if excluded.

The central identification concern is macroeconomic-regime heterogeneity, especially in monetary policy. Although I control for realised exchange-rate movements and policy rates, these controls are imperfect and may themselves respond endogenously to oil shocks. To reduce this source of cross-country heterogeneity, I restrict the sample to Eurozone countries following the adoption of the euro.

Throughout, ξ_m is lagged by four quarters so that it reflects pre-shock production networks and CPI weights, as discussed in Section 5. Standard errors are double-clustered by country and quarter to allow for serial dependence within countries and cross-sectional dependence.

6.1 Main Specifications

6.1.1 Baseline Specification

Following the theoretical framework developed in Section 3, suppose that the underlying data-generating process (DGP) is:

$$\pi_{m,t} = \xi_{m,t-4}u_t + z_{m,t} + \mathbf{W}_{m,t}^\top \boldsymbol{\gamma} + \varepsilon_{m,t}$$

where

$$z_{m,t} \equiv \mathbf{w}_{N,m,t-4}^\top (\mathbf{I} - \mathbf{A}_{NN,m,t-4}^\top)^{-1} (\text{diag}(\mathbf{s}_{N,m,t-4}) \hat{\mathbf{c}}_{N,m,t})$$

denotes the value-added object in Section 3.5 and $\mathbf{W}_{m,t}$ collects other determinants of inflation.

Consistent with the theory, u_t affects CPI inflation exclusively through its interaction with ξ_m , entering with no direct additive effects.

In practice, I omit the value-added term $z_{m,t}$ as it is not directly observed in the data. This omission is not innocuous. If innovations in $z_{m,t}$ are correlated with $\xi_{m,t-4}u_t$, then the estimated coefficient on the interaction term will be biased. However, conditional on lagged inflation, lagged interacted shocks, macroeconomic controls and fixed effects, the remaining variation in $z_{m,t}$ is plausibly uncorrelated with $\xi_{m,t-4}u_t$. I later assess the sensitivity of results to this omission by introducing a proxy for $z_{m,t}$.

Hence, I estimate the following baseline local projection:

$$\Pi_{m,t+h} = \alpha_{m,h} + \beta_h(\xi_{m,t-4}u_t) + \sum_{\ell=1}^3 (\rho_{h,\ell}\pi_{m,t-\ell} + \psi_{h,\ell}(\xi_{m,t-4-\ell}u_{t-\ell}) + \mathbf{x}_{m,t-\ell}^\top \boldsymbol{\gamma}_{h,\ell}) + \varepsilon_{m,t+h}$$

where

$$\Pi_{m,t+h} \equiv \sum_{j=0}^h \pi_{m,t+j}$$

denotes cumulative CPI inflation from quarter t to $t+h$. $\mathbf{x}_{m,t-\ell}$ contains the macroeconomic controls discussed above. $\alpha_{m,h}$ are country fixed effects. Including lagged inflation absorbs persistence in inflation, whereas lagged interacted shocks absorb persistence in exposure-weighted oil shocks. The coefficient of interest is β_h .

6.1.2 With Time Fixed Effects

The baseline specification imposes a strong restriction that oil shocks matter only through ξ_m . More generally, the underlying DGP may instead take the form:

$$\pi_{m,t} = \theta u_t + \beta \xi_{m,t-4}u_t + \phi z_{m,t} + \mathbf{W}_{m,t}^\top \boldsymbol{\gamma} + \varepsilon_{m,t}$$

To allow for this possibility, I also estimate a specification with time fixed effects:

$$\begin{aligned} \Pi_{m,t+h} = & \alpha_{m,h} + \delta_{t,h} + \beta_h(\xi_{m,t-4}u_t) + \sum_{\ell=1}^3 \rho_{h,\ell}\pi_{m,t-\ell} \\ & + \sum_{\ell=1}^3 \psi_{h,\ell}(\xi_{m,t-4-\ell}u_{t-\ell}) + \sum_{\ell=1}^3 \mathbf{x}_{m,t-\ell}^\top \boldsymbol{\gamma}_{h,\ell} + \varepsilon_{m,t+h} \end{aligned}$$

Here, $\delta_{t,h}$ captures all additive shocks common to Eurozone countries in quarter t , including the common oil shock u_t itself. Accordingly, β_h is identified solely from cross-country

differences in ξ within a given quarter.

Thus, this specification asks a more demanding question than the baseline. After removing common quarter-specific disturbances, do countries with higher $\xi_{m,t-4}$ exhibit larger CPI pass-through, weighed against the alternative that oil can affect CPI without interacting with ξ_m .

In the Eurozone sample, as policy rates and exchange rate movements are common across countries in any quarter, they are absorbed by time fixed effects. Therefore, I drop them from the control set to avoid collinearity.

6.2 Additional Specifications

I consider two extensions to my analysis.

First, I augment the baseline specification with the proxy for the value-added term $z_{m,t}$ as mentioned in Section 4.1. This provides a partial check on whether the baseline results are being driven by omission of the value-added channel implied by the model. However, this proxy is a noisy measure of the theoretical object, and the exercise should be interpreted as a sensitivity check rather than a fully structural implementation of the value-added term.

Second, the observed oil shock may not be fully exogenous. Oil price movements may also reflect anticipated changes and global demand conditions (Kilian 2009), which biases results as oil prices and inflation would be simultaneously determined. To address this issue, I re-estimate the baseline specification using two-stage least squares with instrumental variables, using the Känzig (2021) oil supply news shock to construct an instrument for $\xi_{m,t-4}u_t$. The instrument is the corresponding interaction between the news shock series and $\xi_{m,t-4}$. This strategy isolates variation in oil prices driven by exogenous supply news from endogenous movements in global demand.

In the appendix, I also explore second-order effects and compositional effects.

6.3 Robustness Checks

First, I re-estimate both main specifications over alternative time windows: a pre-financial crisis sample (1999–2007) and a pre-pandemic sample (2009–2019). This checks whether the estimated relationship is stable across distinct macroeconomic environments and is not driven by crisis periods.

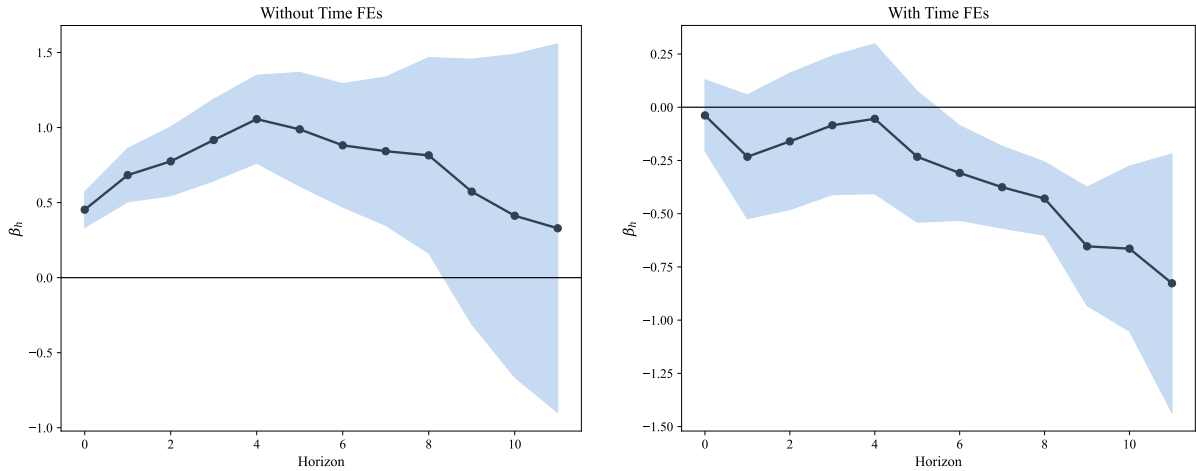
Second, I relax the Eurozone restriction and re-estimate both specifications on a broader country-quarter sample. This examines whether the results generalise beyond the Eurozone. However, the macroeconomic controls are not available for all countries and quarters in the broader sample, with missingness concentrated among developing countries.

7 Results

The central finding is that country-level multipliers ξ_m fail to explain the cross-country heterogeneity observed in CPI inflation. While ξ_m is informative about the average scale of oil shock pass-through, the null result in the time fixed effects specification is robust to alternative specifications and robustness checks.

7.1 Baseline Results

Figure 5 presents the estimated local projections coefficients β_h for $\xi_{m,t-4}u_t$.



Notes: The shaded area represents the 95% confidence interval.

Figure 5: Local projections of cumulative CPI inflation on $\xi_{m,t-4}u_t$

In the baseline specification without time fixed effects, the estimated coefficient path is hump-shaped. The point estimates are positive and statistically significant over short and medium horizons, before flattening out and declining over longer horizons. The estimated full pass-through is 1.056 at $h = 4$, close to unit pass-through predicted by the theory.

Including time fixed effects changes the results. The estimated coefficients are instead negative and statistically insignificant over short and medium horizons, before turning

significantly negative at longer horizons. Countries with larger ξ_m do not exhibit larger CPI pass-through from oil shocks.

7.2 Results from Additional Specifications

The inclusion of additional specifications does not alter the qualitative pattern of the results, especially the null result of the time fixed effects specification.

Figure 6 examines the sensitivity of the baseline results to including the proxy for the omitted value-added term $z_{m,t}$. In both specifications, the pattern remains unchanged.

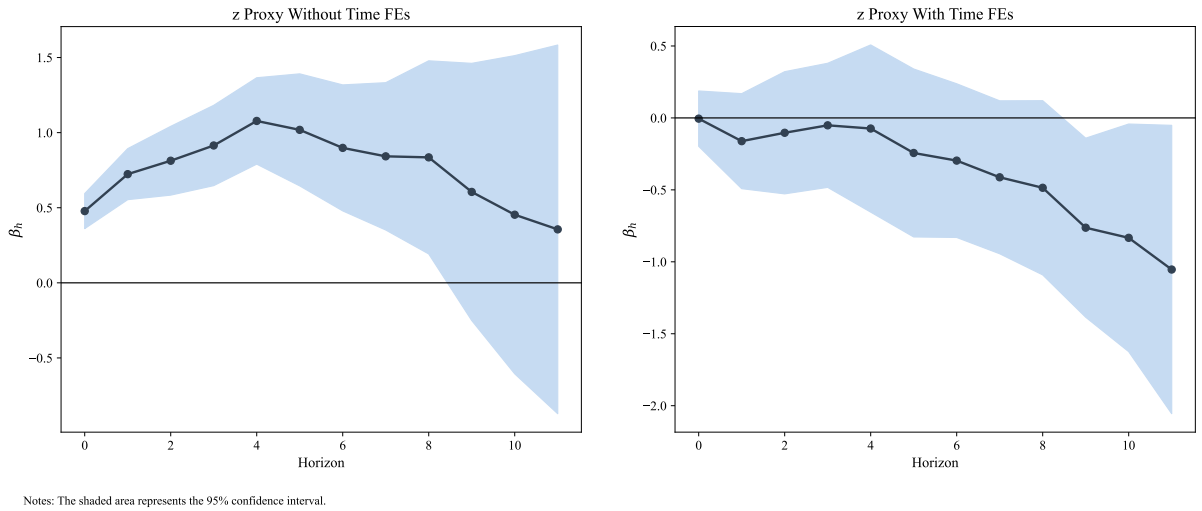


Figure 6: Local projections including a proxy for the value-added term $z_{m,t}$

Figure 7 reports instrumental variables (IV) estimates, where the interaction term between the Känzig (2021) oil supply news shock and $\xi_{m,t-4}$ acts as an instrument for $\xi_{m,t-4}u_t$. Relative to the baseline specification, the IV point estimates are generally similar in sign and shape, but they are much less precisely estimated. For the time fixed effects specification, the point estimates remain close to zero throughout and the negative drift in later horizons exhibited in the main specification is absent.

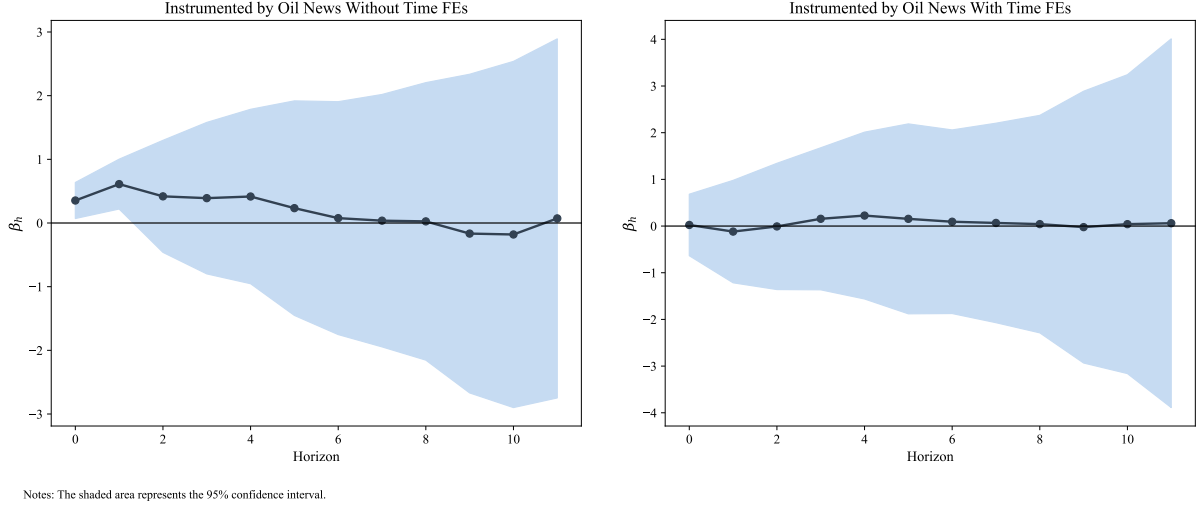


Figure 7: IV local projections

Table 1 shows that without time fixed effects, the first stage regression is strong at the selected horizons, with excluded-instrument F -statistics above 10 throughout. With time fixed effects, F -statistics fall below 10. This is consistent with the much wider confidence intervals in Figure 7, especially in the time fixed effects specification.

Table 1: LP-IV diagnostics at selected horizons

Horizon h	First stage	Excl. IV F	Partial R^2	2SLS $\hat{\beta}_h$	N
<i>Panel A. Without time fixed effects</i>					
0	0.072*** (0.019)	16.47	0.407	0.353** (0.144)	1419
4	0.072*** (0.022)	14.48	0.405	0.416 (0.698)	1343
8	0.075*** (0.022)	14.35	0.429	0.025 (1.111)	1267
<i>Panel B. With time fixed effects</i>					
0	0.042* (0.025)	10.53	0.280	0.024 (0.336)	1419
4	0.035 (0.031)	7.54	0.271	0.225 (0.911)	1343
8	0.038 (0.034)	7.85	0.305	0.042 (1.187)	1267

Notes: “First stage” reports the coefficient on the excluded instrument in the first-stage regression. “Excl. IV F ” reports the excluded-instrument first-stage F -statistic. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.3 Robustness Checks

The main null result is also robust to different sample selections.

Figure 8 examines the robustness of the negative longer-horizon coefficients in the full Eurozone sample time fixed effects specification. The left panel reports estimates for the pre-financial crisis period (1999–2007), while the right panel reports estimates for the pre-pandemic period (2009–2019). The negative pattern seen in Figure 5 is not robust across these subsamples. This suggests the negative results may be driven by large global shocks, most plausibly by the post-pandemic inflation surge. In the specification without time fixed effects, point estimates and confidence intervals are stable across subsamples.

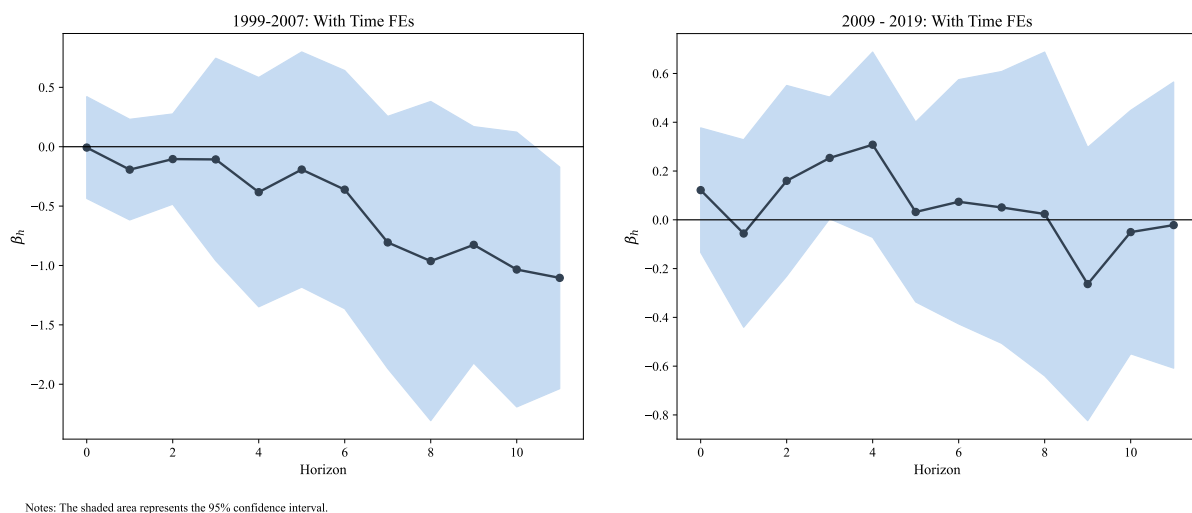


Figure 8: Time fixed effects estimates across alternative subsamples

Figure 9 examines the robustness of the main results by relaxing the Eurozone restriction and estimating the main specifications on the broader country-quarter sample.

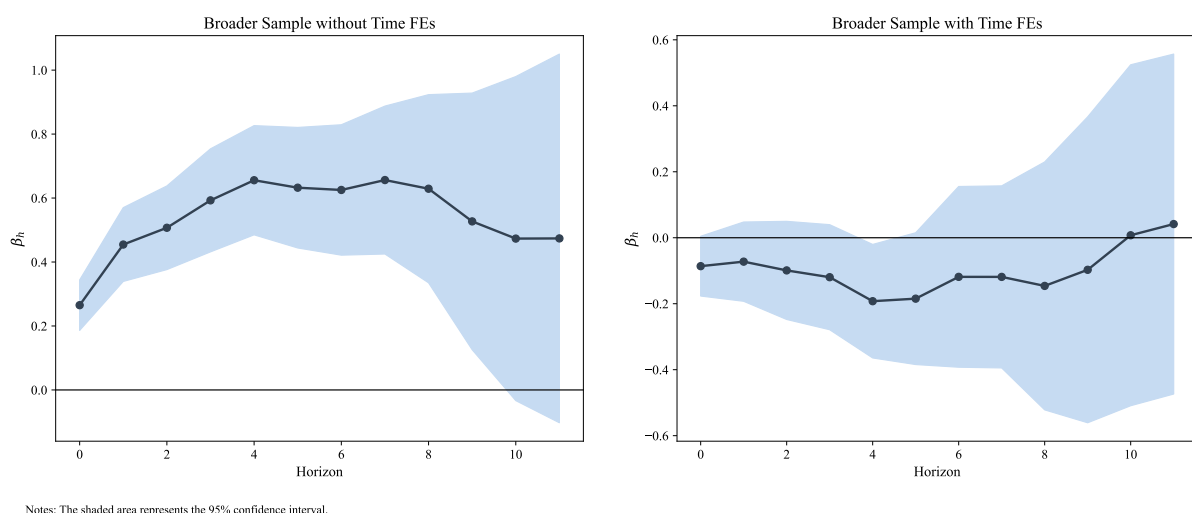


Figure 9: Local projections using the broader country-quarter sample

For the specification without time fixed effects, the estimated coefficients remain positive and the hump-shape persists, so the qualitative response profile is preserved. However, the

magnitude of the pass-through is lower than in the Eurozone baseline. In the specification with time fixed effects, the estimated coefficients remain statistically insignificant and no longer reproduce the negative longer-horizon pattern seen in Figure 5.

8 Discussion and Limitations

This section interprets the central finding that ξ_m does not robustly predict cross-country heterogeneity in CPI responses as theory dictates. I first rule out less plausible explanations and then turn to measurement and theoretical limitations. Finally, I discuss possible future extensions to my current analysis.

In the baseline specification without time fixed effects, the coefficient path is positive, hump-shaped, and close to unity at intermediate horizons. This gives partial support for the economic significance of the country-level multiplier ξ . However, the heterogeneity null in the time fixed effects specification is robust across every extension: instrumental variables, a value-added proxy, subsample splits, and a broader country panel.

Several candidate explanations can be ruled out. First, monetary regime heterogeneity cannot be the primary driver, because the null holds within the Eurozone. Second, the positive results in the specification without time fixed effects and existing literature (Choi et al. 2018) confirm that oil shocks do pass through into inflation. So pass-through is not substantially muted by policy responses; the finding is the absence of differential pass-through, not absence of pass-through altogether. Third, large-scale fiscal responses are also unlikely. The null persists even after excluding the Great Financial Crisis and the COVID-19 pandemic. Similarly, government interventions in retail energy pricing such as price caps or fuel subsidies are concentrated in crisis periods and cannot explain a null that survives in subsamples. Separately, permanent cross-country differences in product-tax regimes are also absorbed by country fixed effects.

The most parsimonious econometric explanation for the null is attenuation bias. ξ is constructed from annual ICIO tables and HFCE shares, then matched to quarterly CPI data. The mapping from ξ to inflation is indirect, aggregating across both sectors and countries, which makes it prone to measurement error. Once time fixed effects are introduced, identification comes solely from cross-sectional variation. Measurement error relative to the quarterly noise in observed CPI may be large enough to bias the coefficients towards zero.

On the theory side, the Leontief benchmark also assumes full, immediate cost pass-through, but pass-through could be incomplete and delayed. First, sectors differ in price

rigidity. Sectors such as food production transmit costs within weeks, while sectors such as administrative services may take several quarters to adjust. Without accounting for price rigidity, ξ fails to capture this cross-country heterogeneity. Second, the zero-profit assumption rules out markup absorption. Firms may absorb part of the cost shock in margins. The degree of absorption varies across countries depending on market structure. Third, in an integrated goods market, some tradeable sectors are effectively exogenous, with prices set on the whole European market. Extending the analysis to the broader sample does not fully resolve this issue. Even outside a formal single market, tradeable goods prices are partially disciplined by international competition, even if not fully determined by it.

These mechanisms are not mutually exclusive. Attenuation bias compresses the point estimates towards zero, while incomplete pass-through both weakens the underlying relationship and adds noise orthogonal to ξ .

The empirical analysis is limited by the lack of sectoral price indices, which forces the test to headline CPI. Actual CPI weights also differ from HFCE shares used to proxy them, compounding measurement error in ξ_m . On the theoretical side, the natural extension is the microfounded CES model in the appendix, but exploiting its higher order terms empirically requires richer data than is currently available. More promisingly, an enhanced ξ that weights sectoral exposure by price rigidity would sharpen the cross-country heterogeneity test.

9 Conclusion

The principal contribution of this dissertation is the construction of theoretically motivated and economically interpretable measures of embedded oil exposure at the sectoral level $\tau_{N,m}$, and the country level ξ_m . Empirically, the baseline specification suggests that ξ_m captures the scale of the average pass-through. However, I find little evidence that the country-level multiplier ξ_m robustly explains within-quarter cross-country differences in CPI pass-through, suggesting that there are important mechanisms neglected by the baseline theoretical framework.

A substantial limitation is that the empirical analysis had to work with highly aggregated prices, as the mapping from sectoral cost exposure to CPI is obscured by many factors, including endogenous substitution, markup absorption, and heterogeneous sectoral price rigidities. Future work with sectoral price data would permit a more granular test of the relationship between production network structures and shock pass-through.

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A CES Microfoundation

Consider sector $i \in N$. Let u denote a scalar exogenous shock, and treat prices as functions of u . Under constant returns to scale, cost minimisation, and perfect competition, the dual unit cost function under CES is

$$p_i(u) = \left[\sum_{j \in N} \omega_{ji} p_j(u)^{1-\sigma_i} + \sum_{e \in E} \omega_{ei} p_e(u)^{1-\sigma_i} + \omega_{vi} c_i(u)^{1-\sigma_i} \right]^{\frac{1}{1-\sigma_i}}$$

where $\sigma_i > 0$ is the elasticity of substitution and

$$\sum_{j \in N} \omega_{ji} + \sum_{e \in E} \omega_{ei} + \omega_{vi} = 1$$

Define for any variable x :

$$\tilde{x}(u) \equiv \frac{\partial \ln x(u)}{\partial u}$$

By Shephard's lemma, the corresponding cost shares are

$$a_{ji}(u) = \omega_{ji} \left(\frac{p_j(u)}{p_i(u)} \right)^{1-\sigma_i} \quad a_{ei}(u) = \omega_{ei} \left(\frac{p_e(u)}{p_i(u)} \right)^{1-\sigma_i} \quad a_{vi}(u) = \omega_{vi} \left(\frac{c_i(u)}{p_i(u)} \right)^{1-\sigma_i}$$

with

$$\sum_{j \in N} a_{ji}(u) + \sum_{e \in E} a_{ei}(u) + a_{vi}(u) = 1$$

Differentiating the unit cost function gives the first-order price system

$$\tilde{p}_i(u) = \sum_{j \in N} a_{ji}(u) \tilde{p}_j(u) + \sum_{e \in E} a_{ei}(u) \tilde{p}_e(u) + a_{vi}(u) \tilde{c}_i(u)$$

As in Section 3.4, $\tilde{p}_e(u) = \lambda_e$

Then

$$\tilde{p}_i(u) = \sum_{j \in N} a_{ji}(u) \tilde{p}_j(u) + \sum_{e \in E} a_{ei}(u) \lambda_e + a_{vi}(u) \tilde{c}_i(u)$$

In vector form,

$$\tilde{\mathbf{p}}_N(u) = \mathbf{A}_{NN}(u)^\top \tilde{\mathbf{p}}_N(u) + \mathbf{A}_{EN}(u)^\top \boldsymbol{\lambda} + \text{diag}(\mathbf{a}_v(u)) \tilde{\mathbf{c}}_N(u)$$

Note that $\mathbf{a}_v(u)$ is equivalent to $\mathbf{s}_N$ as in Section 3.3.

Evaluate at the benchmark $u = 0$, and denote benchmark objects by bars. Then

$$\tilde{\mathbf{p}}_N = \bar{\mathbf{A}}_{NN}^\top \tilde{\mathbf{p}}_N + \bar{\mathbf{A}}_{EN}^\top \boldsymbol{\lambda} + \text{diag}(\bar{\mathbf{a}}_v) \tilde{\mathbf{c}}_N,$$

so

$$\tilde{\mathbf{p}}_N = (\mathbf{I} - \bar{\mathbf{A}}_{NN}^\top)^{-1} \bar{\mathbf{A}}_{EN}^\top \boldsymbol{\lambda} + (\mathbf{I} - \bar{\mathbf{A}}_{NN}^\top)^{-1} \text{diag}(\bar{\mathbf{a}}_v) \tilde{\mathbf{c}}_N.$$

Here, $\boldsymbol{\tau} \equiv (\mathbf{I} - \bar{\mathbf{A}}_{NN}^\top)^{-1} \bar{\mathbf{A}}_{EN}^\top \boldsymbol{\lambda}$ captures the first-order response of a sector's log price to the common exogenous shock.

As in Section 3.5, ξ is recovered by aggregating sectoral prices to consumer prices using consumption expenditure weights.

To obtain second-order effects, differentiating once more with respect to u gives

$$\tilde{p}'_i(u) = \sum_{j \in N} \left[a'_{ji}(u) \tilde{p}_j(u) + a_{ji}(u) \tilde{p}'_j(u) \right] + \sum_{e \in E} \left[a'_{ei}(u) \tilde{p}_e(u) + a_{ei}(u) \tilde{p}'_e(u) \right] + a'_{vi}(u) \tilde{c}_i(u) + a_{vi}(u) \tilde{c}'_i(u)$$

$a'_{ji}(u)$, $a'_{ei}(u)$, and $a'_{vi}(u)$ capture endogenous changes in cost shares as relative input prices move.

Collecting the reallocation terms into $\Gamma_i(u)$:

$$\tilde{p}'_i(u) = \sum_{j \in N} a_{ji}(u) \tilde{p}'_j(u) + \sum_{e \in E} a_{ei}(u) \tilde{p}'_e(u) + a_{vi}(u) \tilde{c}'_i(u) + \Gamma_i(u)$$

where

$$\Gamma_i(u) = (1 - \sigma_i) \left[\sum_{j \in N} a_{ji}(u) \tilde{p}_j(u)^2 + \sum_{e \in E} a_{ei}(u) \tilde{p}_e(u)^2 + a_{vi}(u) \tilde{c}_i(u)^2 - \tilde{p}_i(u)^2 \right]$$

The term $\Gamma_i(u)$ captures the second-order effects generated by endogenous cost-share adjustment. Its magnitude increases with the dispersion of input price responses, weighted by cost shares. Its sign is determined by σ_i . If $0 < \sigma_i < 1$, inputs are gross complements and pass-through is amplified; if $\sigma_i > 1$, substitution dampens it.

Energy is plausibly a gross complement in the short run, since firms cannot easily switch fuel sources or redesign processes within quarters. Over longer horizons, substitution becomes feasible through fuel switching or energy-savings investments.

B Extensions: Nonlinearities and Composition

I estimate two exploratory extensions to the baseline specification.

First, I test for second-order nonlinearities by augmenting the baseline specification with a quadratic oil shock term and its lags. The quadratic oil shock term is common across countries within a quarter, so this specification is estimated without time fixed effects.

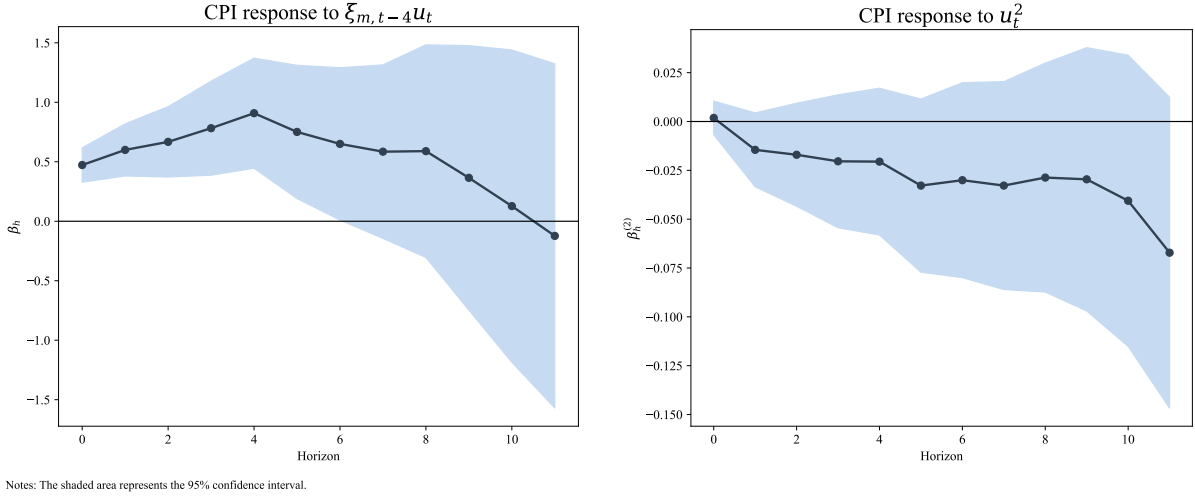


Figure 10: Local projections with a quadratic oil-shock term

Figure 10 reports the resulting coefficient paths. The coefficient on the interaction term remains similar to the baseline, while the quadratic term is small and imprecisely estimated throughout. There is little evidence that second-order nonlinearities materially alter the first-order pass-through relationship.

Second, I examine whether the composition of ξ_m matters beyond its total level by replacing $\xi_{m,t-4}u_t$ with separate interactions for $\xi_{\text{dir},m}$, $\xi_{\text{near},m}$, and $\xi_{\text{deep},m}$. This specification omits time fixed effects.

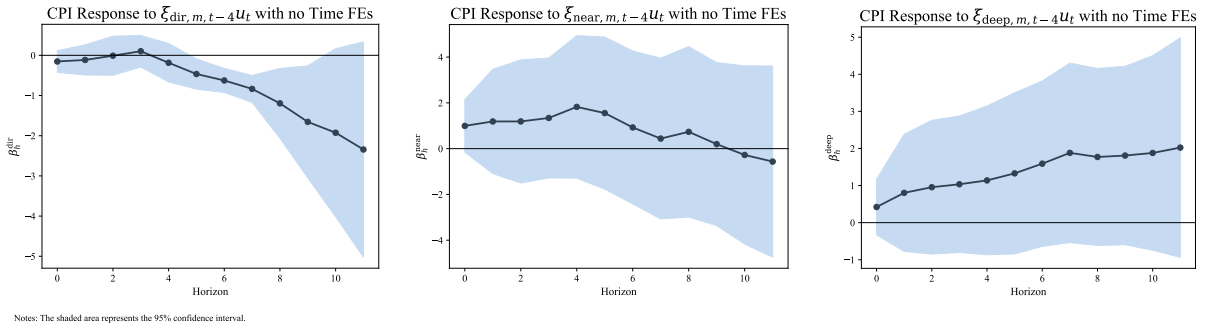


Figure 11: Local projections by components of the country-level multiplier ξ

The point estimates suggest that the deep component is associated with a more persistent CPI response than the near component. However, the standard errors of all three components are large. These results are therefore descriptive rather than conclusive.